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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

eXTRA

SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Extragorjoss
Selected Domain	2
Selected SDG Track (1–17)	8.2 & 8.3
Project / Solution Name	CashGuard Risk (An actuarial cash-safety check for micro and small businesses)
Name (Leader)	MUHAMMAD AMEER AMSYAR BIN HAMDAN
Email & Phone (Leader)	ameer.amsyar07@gmail.com · 019-441 2080
University Affiliation(s)	Universiti Teknologi MARA (UiTM); Universiti Teknikal Malaysia Melaka (UTeM)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Micro and small businesses like local cafés, workshops, and retail shops face a severe daily challenge: managing cash flow safely. Most owners make spending decisions by simply checking their current bank balance. However, this balance is misleading because it ignores upcoming expenses like rent or payroll, as well as customers who pay late. Paying a supplier bill today can easily leave a business without enough money to cover basic costs next week.

This problem directly affects small business owners, their staff, and local suppliers. Small enterprises are the backbone of local communities, providing vital jobs and services. When cash runs out, owners face extreme stress, employees risk delayed wages, and suppliers go unpaid.

The economic and social consequences are far-reaching. Cash shortfalls can push small businesses into debt or sudden failure, disrupting local supply chains and causing immediate job losses that hurt families. Furthermore, without clear visibility into customer payment habits, owners cannot plan spending or protect their financial safety net.

Solving this problem aligns with UN Sustainable Development Goals 8.2 and 8.3 by using practical technology to support small enterprises. Giving owners simple, clear insights into cash risks before they pay bills protects business stability, keeps workers employed, and strengthens local communities.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

CashGuard is a cash-safety tool that helps small business owners decide when it is safe to pay a bill or approve spending. Instead of relying on today's bank balance, it looks ahead 30 days. For every supplier invoice it calculates the chance that paying today leaves the business below its cash reserve, suggests a safer payment date that is still on time, and explains why in plain language. The owner always decides, and every decision is logged.

Invoices arrive through a WhatsApp bot. Suppliers send a photo of their invoice, which document AI (an open-source vision model) reads; or, if they have no invoice, they fill in a WhatsApp form and the bot generates the invoice for them with default details. The bot then keeps suppliers updated: received, approved with the transfer date, and paid. Checks catch maths errors, duplicates and changed bank details. Survival-analysis models predict when customers will really pay, and 1,000 simulated cash futures measure the risk. Spending limits for staff and managers are set by the owner. AI never approves or makes a payment.

From the Phase 1 workshops we apply: GenAI APIs (a hosted vision-language model called through an OpenAI-compatible API to extract invoice data), database architecture (a PostgreSQL schema with row-level security and an insert-only audit log), and cloud deployment (the web app and WhatsApp bot hosted in the cloud with keys kept in secrets, and the AI model on a GPU cloud).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Front end: Streamlit (Python) web app with role-based screens for Owner, Manager and Employee; Plotly charts. The owner sets manager approval limits (per item and per month) and staff spending categories.

Input channel: WhatsApp bot. A Flask webhook (Twilio WhatsApp API or Meta WhatsApp Cloud API) receives invoice photos, or a fill-in form from suppliers with no invoice and generates the invoice PDF (ReportLab) with default details. It checks the sender, duplicates and changed bank details, and updates suppliers when their invoice is received, approved with a pay date, and paid. Suppliers join by a link or QR code.

GenAI / document AI: GLM-OCR (open-source vision model, MIT licence) served with vLLM through an OpenAI-compatible API. Returns invoice fields as JSON; a fallback to manual entry keeps the app working if the model is down.

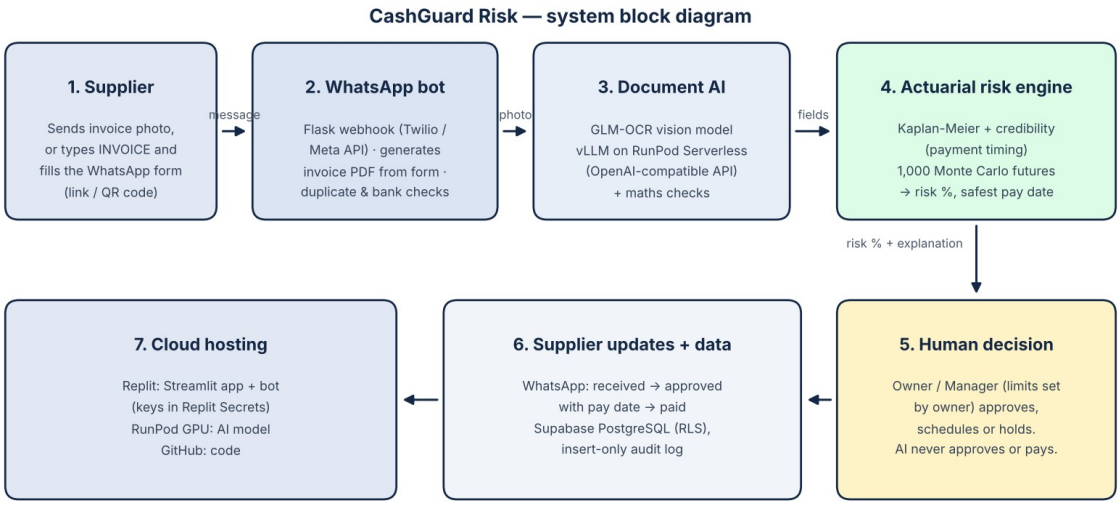
Analytics engine: Python, NumPy and pandas. Kaplan-Meier survival analysis with Bühlmann credibility for customer payment timing, and a 1,000-path Monte Carlo cash simulation for risk %, best payment date and plain-language explanation.

Database: Supabase PostgreSQL with Row Level Security, Supabase Auth, a private storage bucket for invoice files, and an insert-only audit log.

Cloud hosting: Replit hosts the web app and webhook (keys in Replit Secrets); RunPod GPU hosts the AI model; code on GitHub.

How it connects: WhatsApp / upload → webhook + web app → document AI (RunPod) → staff confirm → risk engine → owner or manager decides → Supabase database and audit log; all hosted on Replit. See the block diagram below.

Sponsor resources: Yes. Replit (cloud platform) hosts the live prototype: the Streamlit app and the WhatsApp webhook. RunPod (GPU cloud) runs the GLM-OCR model that reads invoices. Neither is a single point of failure: the risk engine is our own Python code and runs without any sponsor service.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

To build a practical solution, our team applied design-thinking methodology centered around the needs of small business owners.

During empathy and problem definition, we observed that micro-enterprises rely heavily on paper records, WhatsApp, and basic spreadsheets. Owners lack formal finance training and feel overwhelmed by complex data, while employees need simple ways to submit expenses without viewing sensitive cash metrics.

To ensure user-friendliness, we ideated a simple role-based dashboard. Employees easily upload invoices while AI pre-fills the data to eliminate manual entry. For owners, we replaced complex financial jargon with plain-language explanations, clear percentages, and visual color-coded risk indicators.

For technical viability and deployment readiness, we designed a hybrid architecture. We combined open-source document AI for invoice reading with an explainable actuarial risk engine that runs Monte Carlo simulations. Synthetic demo data validated our predictive accuracy before using real financial records. Finally, we implemented cloud hosting, secure database access, and an append-only audit log to guarantee data privacy, safety, and seamless real-world adoption by Phase 2.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 8.2: "Achieve higher levels of economic productivity through diversification, technological upgrading and innovation, including through a focus on high-value added and labour-intensive sectors." Target 8.3: "Promote development-oriented policies that support productive activities, decent job creation, entrepreneurship, creativity and innovation, and encourage the formalization and growth of micro-, small- and medium-sized enterprises, including through access to financial services."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	**Real-world context:** MSMEs provide 48.7% of Malaysia's employment (8.10 million people) and 39.5% of GDP [1]. Malaysian SMEs wait an average of 64 days to be paid, versus 58 days for large corporations [2], yet most owners decide what to pay by looking at today's bank balance. **Measured baseline (our benchmark):** On 50 bill situations for a typical micro café (synthetic, fictional data), paying each bill on arrival leaves cash below the RM2,000 safety reserve in 14.4% of simulated futures. Assuming customers pay on the due date mispredicts payment timing by 6.2 days on average.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption,*	**1. Cash shortfall rate (%):** share of futures where cash falls below the business's minimum reserve, judged on futures the model did not see.

Metric Category	Your Response
processing-latency improvement, or kg of resource waste avoided.	**2. Payment-timing error (days):** mean absolute error between predicted and actual customer payment dates (backtest on newer invoices). **3. On-time bill payment (%):** share of bills still paid by their due date when following CashGuard's suggested date.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	**1. ≥75% fewer cash shortfalls:** from 14.4% to 3.5% or lower. The current prototype reaches 3.3% (77% fewer); e.g. a RM1,800 supplier bill falls from 23.7% risk if paid today to 2.1% if paid 4 days later, still before its due date. **2. 29% lower payment-timing error:** from 6.2 days to 4.4 days. **3. 100% of bills paid on time:** safer timing must never make a bill late (prototype: 50 of 50). **Sources:** [1] DOSM (31 Jul 2025), Micro, Small & Medium Enterprises (MSMEs) Performance 2024, https://www.dosm.gov.my/portal-main/release-content/micro-small--medium-enterprises-msmes-performance-2024 [2] Experian Malaysia (21 May 2025), State of Credit 2025, https://www.experian.com.my/2025/05/21/experian-malaysias-state-of-credit-2025-reveals-sme-resilience-but-cash-flow-strains-persist/

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [ ] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [ ] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [ ] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ /No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python 3.13, Streamlit, pandas, NumPy, Plotly	Code (open-source libraries)	PSF, Apache 2.0, BSD-3, MIT
Flask, Requests, Pillow, ReportLab, segno	Code (open-source libraries)	BSD-3, Apache 2.0, MIT-CMU, BSD, BSD-3
GLM-OCR (Z.ai, 0.9B) served with vLLM	Pre-trained model + serving code	MIT; vLLM Apache 2.0
Twilio / Meta WhatsApp APIs	Third-party API (messaging)	Provider terms of service
Supabase (PostgreSQL)	Database platform	Apache 2.0 / PostgreSQL licence
Demo data (business, invoices, sales)	Dataset, synthetic, created by our team	Our own work (no personal data)

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐/ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

25 October 2026